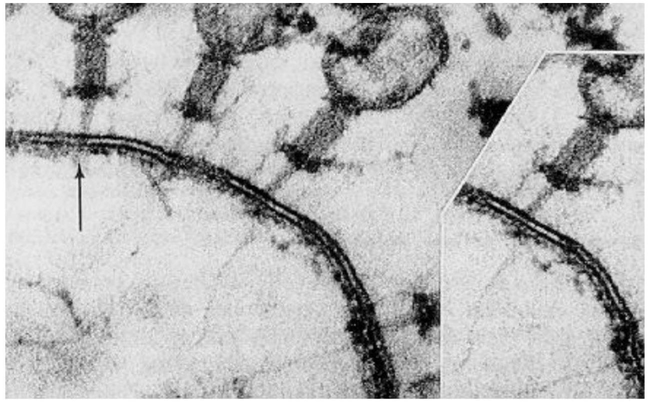


Figure 1. Phage cycle

Source: (microBIO, 2012)

UNOFFICIAL ENGLISH TRANSLATION

Bacteriophage Therapy | Rediscovering an Innovative Therapy

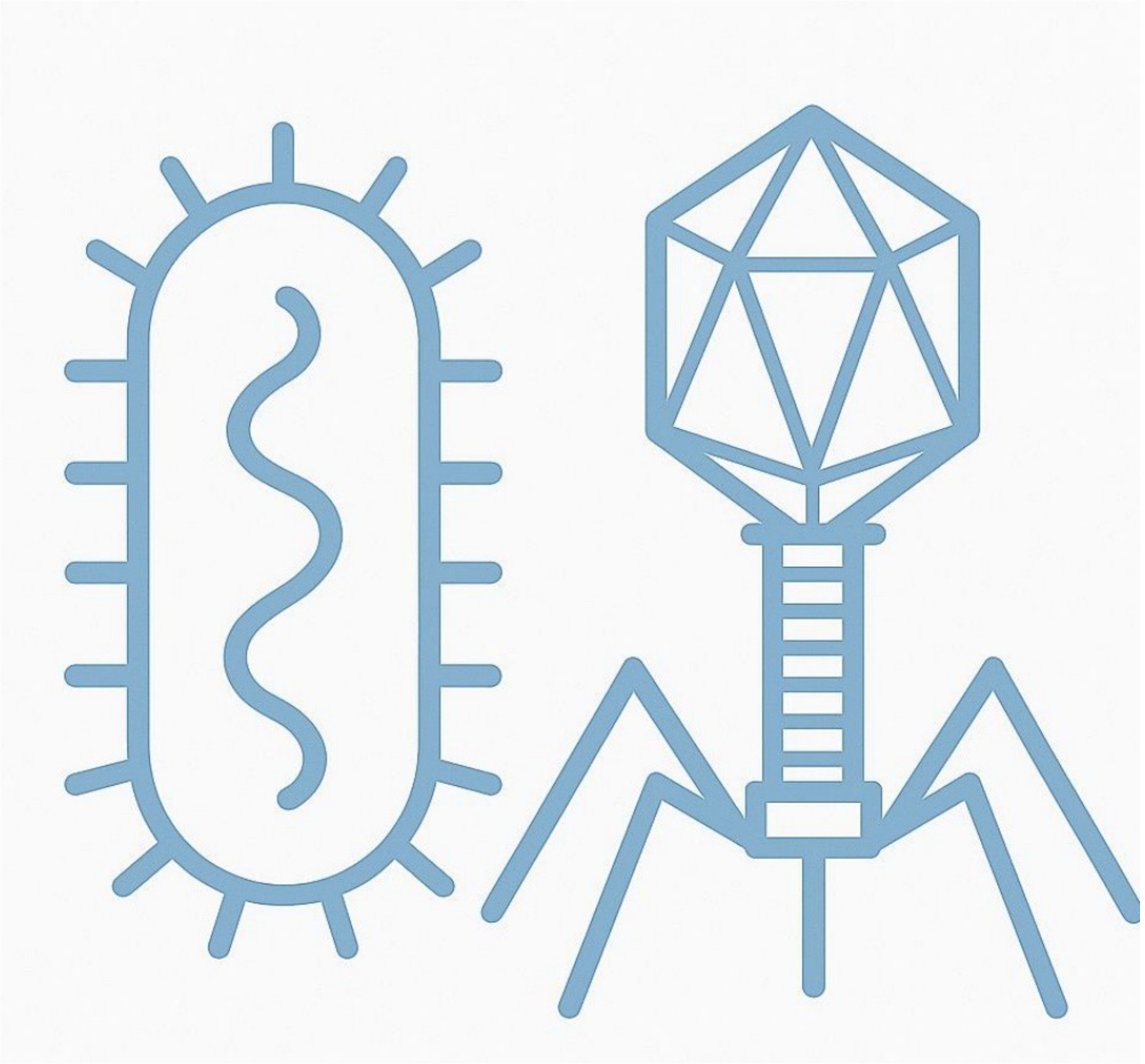
Jhon Michael Cuenca Jaramillo & Luis Gonzalo Legua Pérez

Academic year 2024–2025

Unofficial English translation of the original Spanish defense slides.

Contents

1. The problem
2. What is a bacteriophage?
3. How does a phage act?
4. What is bacteriophage therapy?
5. Advantages over antibiotics
6. Barriers and limitations
7. Clinical case 1
8. Clinical case 2
9. Innovative proposal
10. Conclusion



“Bacterial resistance has surpassed our conventional defenses. It is time to act with new solutions before the irreversible becomes everyday.”

Adapted from the World Health Organization (WHO)

The problem

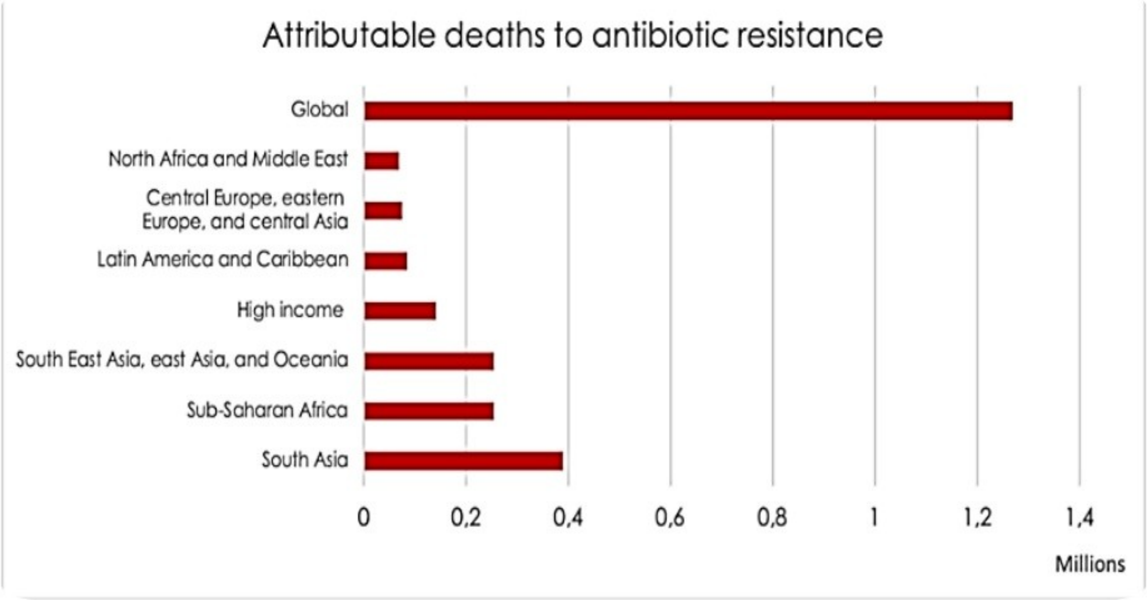


Chart 1. Deaths attributed to antibiotic resistance (GRAM study)

Source: (Salud y Fármacos, 2022)

Bacterial resistance is already a real threat to public health, making treatments more difficult and increasing mortality.

As antibiotics become increasingly ineffective, bacteriophage therapy is re-emerging as a promising alternative.

This project explores its potential, mode of action, clinical applications and current challenges.

What is a bacteriophage?

A bacteriophage is a virus specialised in infecting bacteria. (d'Herelle, 1917)

Structurally, it consists of a protein capsid that contains its genetic material, deoxyribonucleic acid or ribonucleic acid (DNA or RNA). Most bacteriophages also have a specialised tail that allows them to bind to specific receptors on the bacterial membrane. (Bacteriófago, 2023)

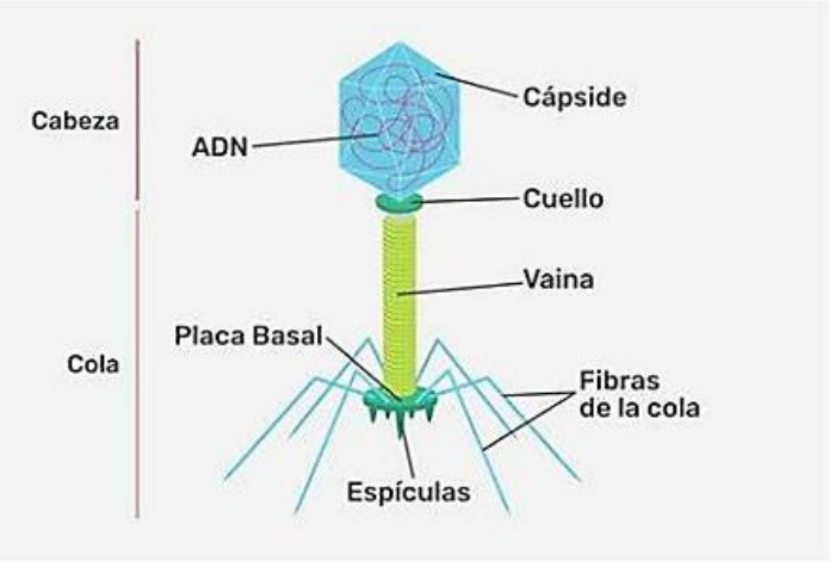


Figure 2. Bacteriophage

Source: (Red de Recursos Educativos en Abierto, n.d.)

How does a phage act?

• **Lytic cycle:** The virus replicates rapidly inside the bacterium, generating virions that ultimately cause cell lysis and release new viruses.

• **Lysogenic cycle:** The phage remains latent inside the bacterial cell, integrating its DNA into the host genome. It can be activated and switch to the lytic cycle after stimuli such as radiation or temperature changes.

• **Pseudolysogenic cycle:** Only some bacteria are susceptible to the phage. It is an intermediate state between the lysogenic and lytic cycles. In chronic infections, some bacteria release new viruses without dying, through budding or extrusion. (da Rosa, 2015)

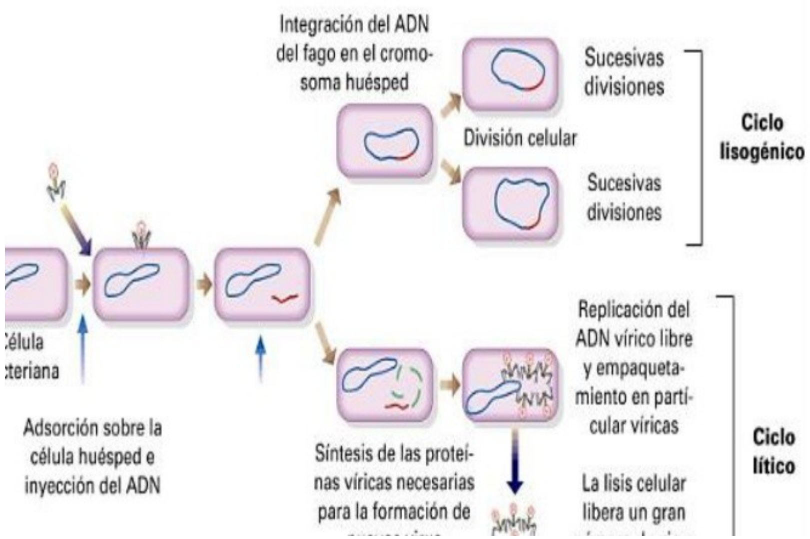


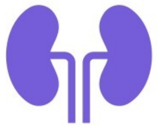
Figure 3. Phage cycle

Source: (Melendez, n.d.)

What is bacteriophage therapy?

A therapeutic strategy that uses bacteriophages to treat bacterial infections, especially those caused by bacteria resistant to antibiotics.

Dr. d'Hérelle's research emphasised that its application may be both preventive and curative, and that it does not affect the patient or damage
any of their cells.



It has a self-regulating function: activity is higher during infection and decreases when the infection begins to disappear from the organism.

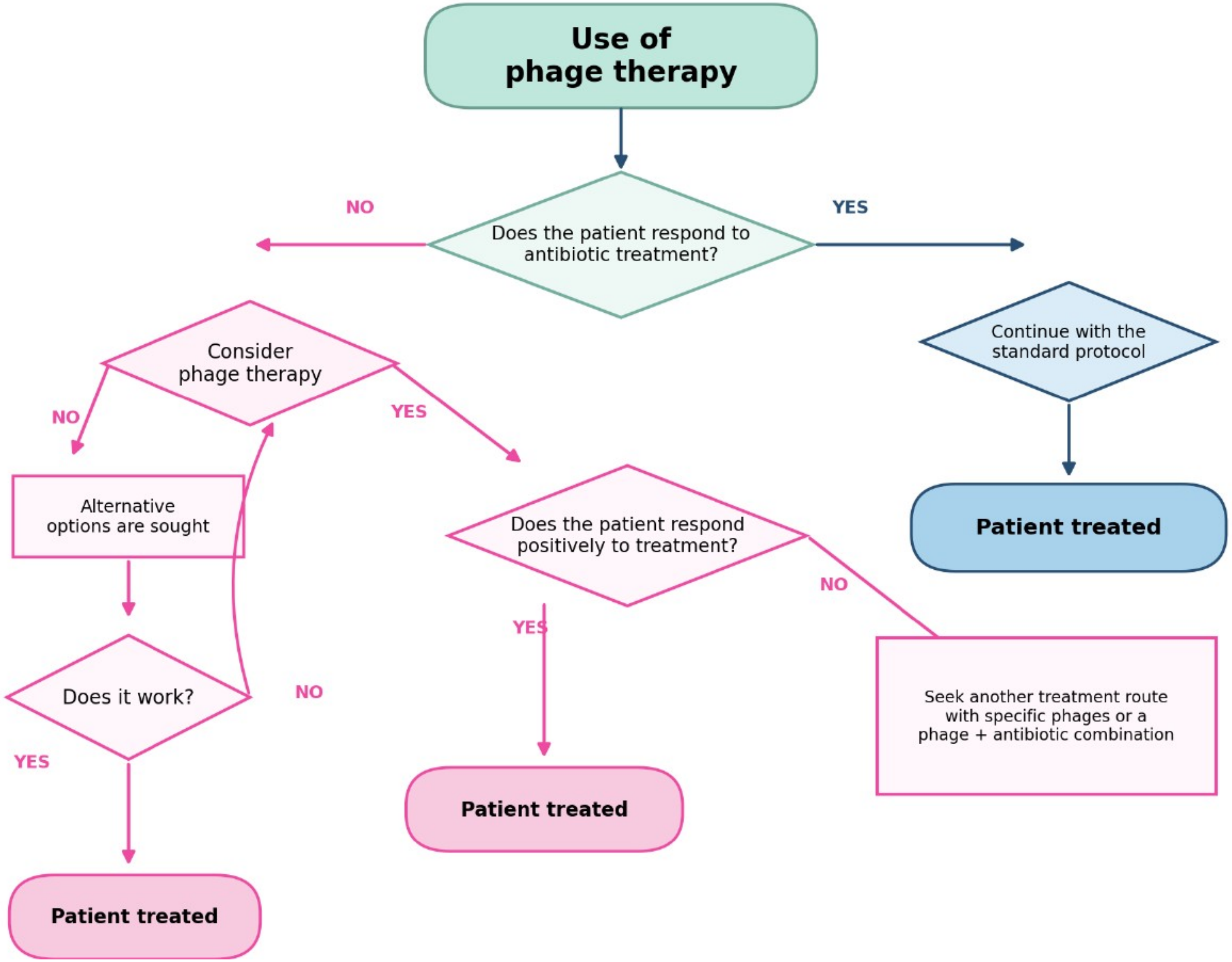


Figure 5. Flowchart on the use of phage therapy

Source: Authors' own work

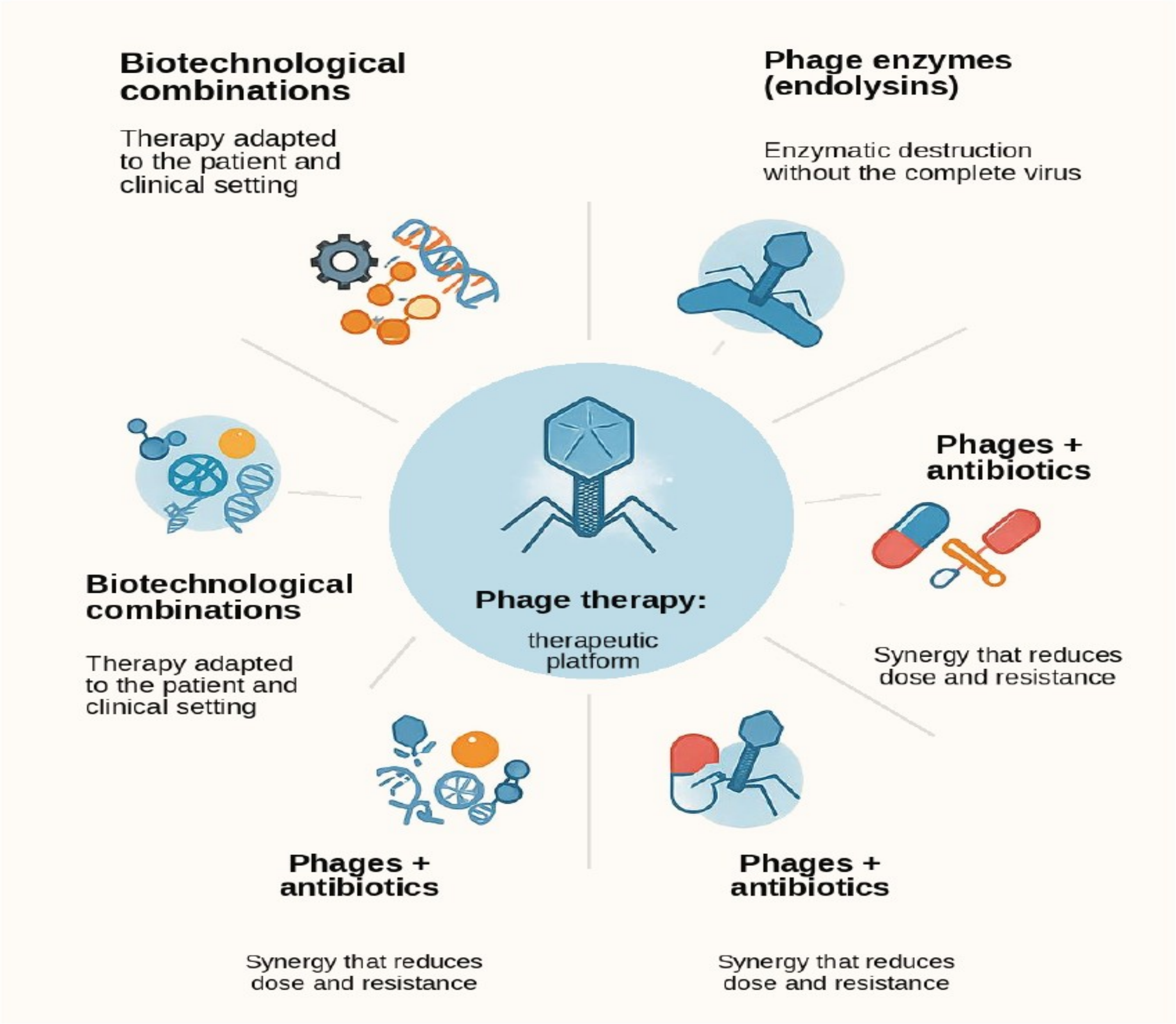


Figure 4. Therapeutic projection of phage therapy in combination with antimicrobial technologies

Source: Authors' own work

Advantages over antibiotics

- High specificity
- Lower impact on microbiota
- Effectiveness against multidrug-resistant bacteria
- Self-replication at the infection site
- Evolutionary adaptation capacity
- Potential for personalised design

Barriers and limitations

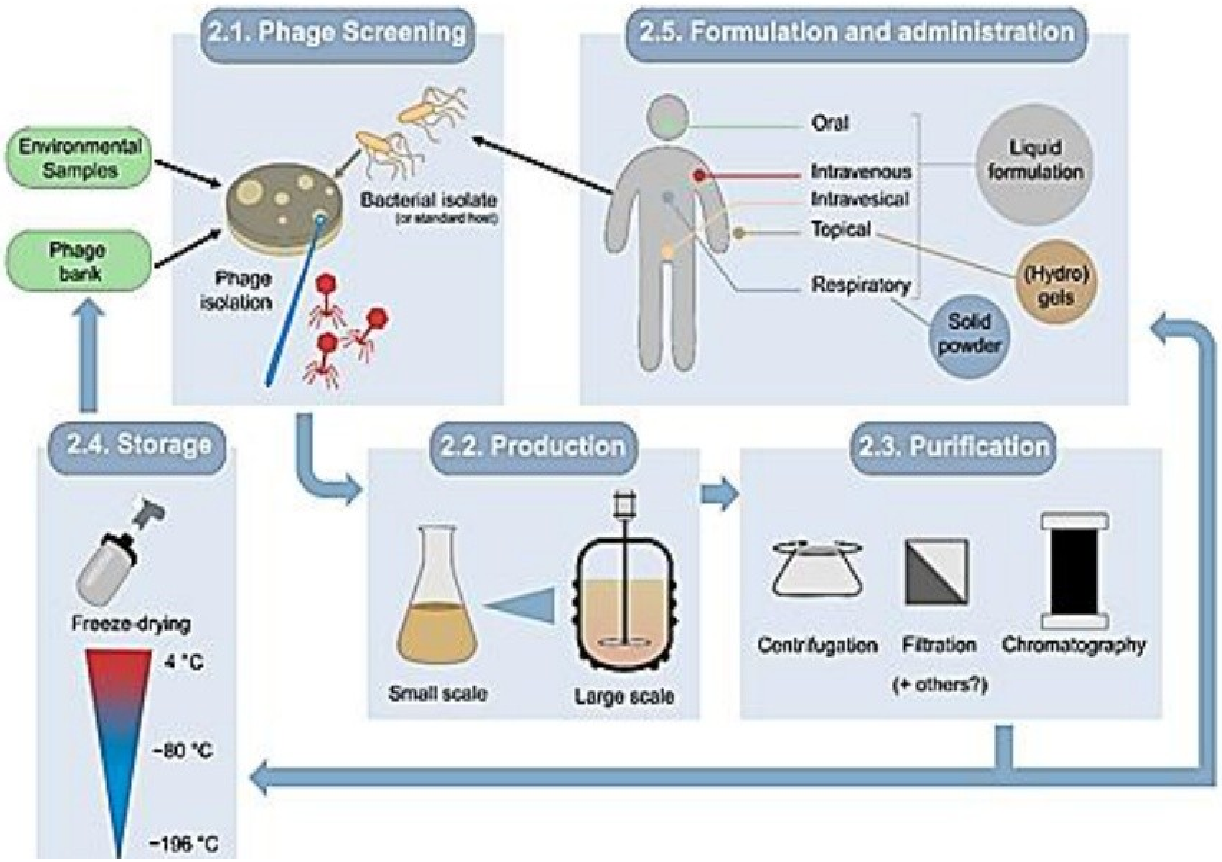


Figure 6. Steps for preparing a phage suspension

Source: (Vázquez et al., 2022)

- High specificity (advantage and limitation)
- Bacterial mutations (phage cocktail)
- Immune response
- Lack of standardised regulation
- High production costs because it is a personalised therapy, although relatively cheaper than other personalised therapies

Clinical case 1

Cystic fibrosis with chronic infections by Staphylococcus aureus and/or Pseudomonas aeruginosa (2024)

This case, which took place in Spain, represents a milestone and was carried out through collaboration between the Institute for Integrative Systems Biology (I2SysBio), the Spanish National Research Council (CSIC) and the University of Valencia (UV), together with the Yale Center for Phage Therapy in the United States.

The treatment was carried out at Hospital Universitario La Fe in Valencia and Hospital Virgen del Rocío in Seville, marking the beginning of the first treatment produced entirely in Spain. It was structured in three stages using three different phage types administered by nebulisation. (CSIC, 2024)

During treatment, the patient was observed to develop antibodies against the phages.

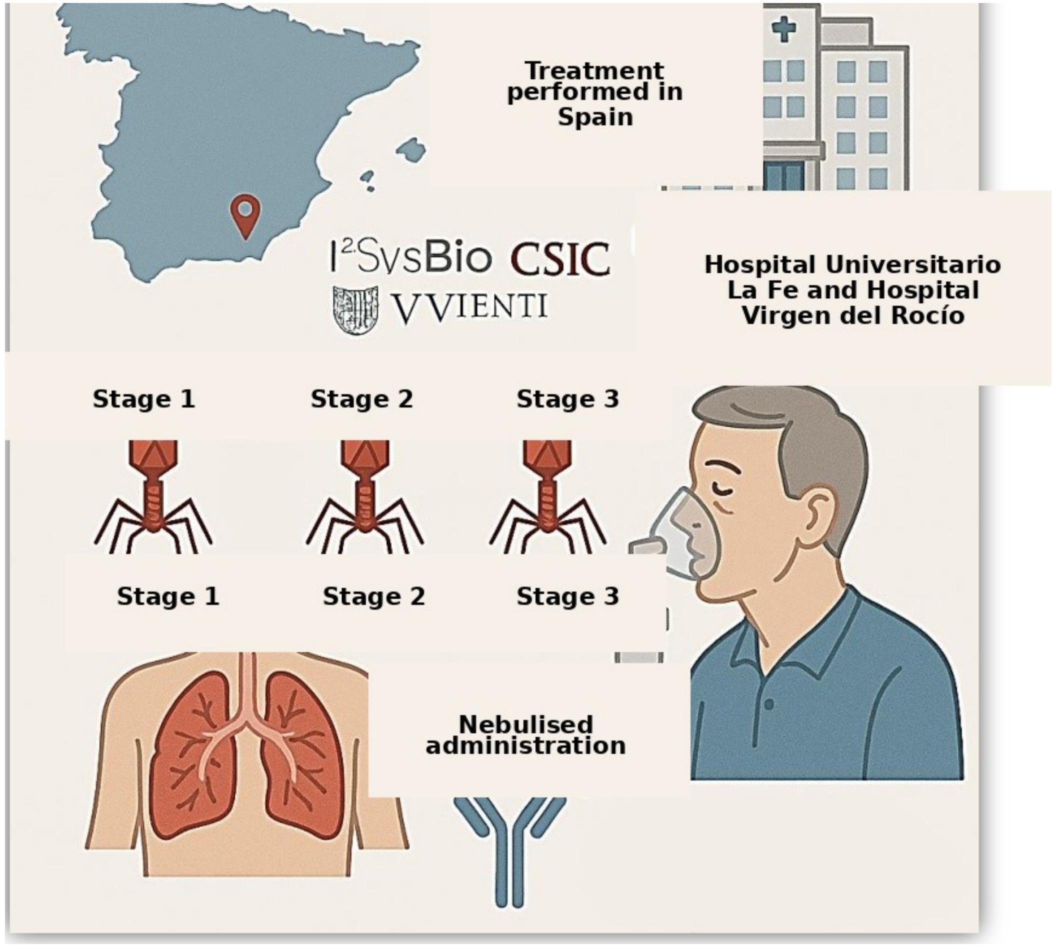


Figure 7. Schematic based on the clinical case

Source: Authors' own work

Clinical case 2

Necrotising pancreatitis complicated by a pancreatic pseudocyst infected with multidrug-resistant Acinetobacter baumannii (2017)

Collaboration among several institutions was key to carrying out this treatment. The patient was treated at UC San Diego Health, while the development and analysis of the phage cocktail were performed by the Center for Phage Technology at Texas A&M University and the US Navy Medical Research Center.

AmpliPhi Biosciences also contributed purified phages for compassionate use, which made it possible to combine intravenous administration with intra-abdominal administration. (Schooley et al., 2017)

The aspect that has generated the greatest clinical interest is the speed of recovery observed after direct administration of the treatment, both intravenously and directly to the affected area.

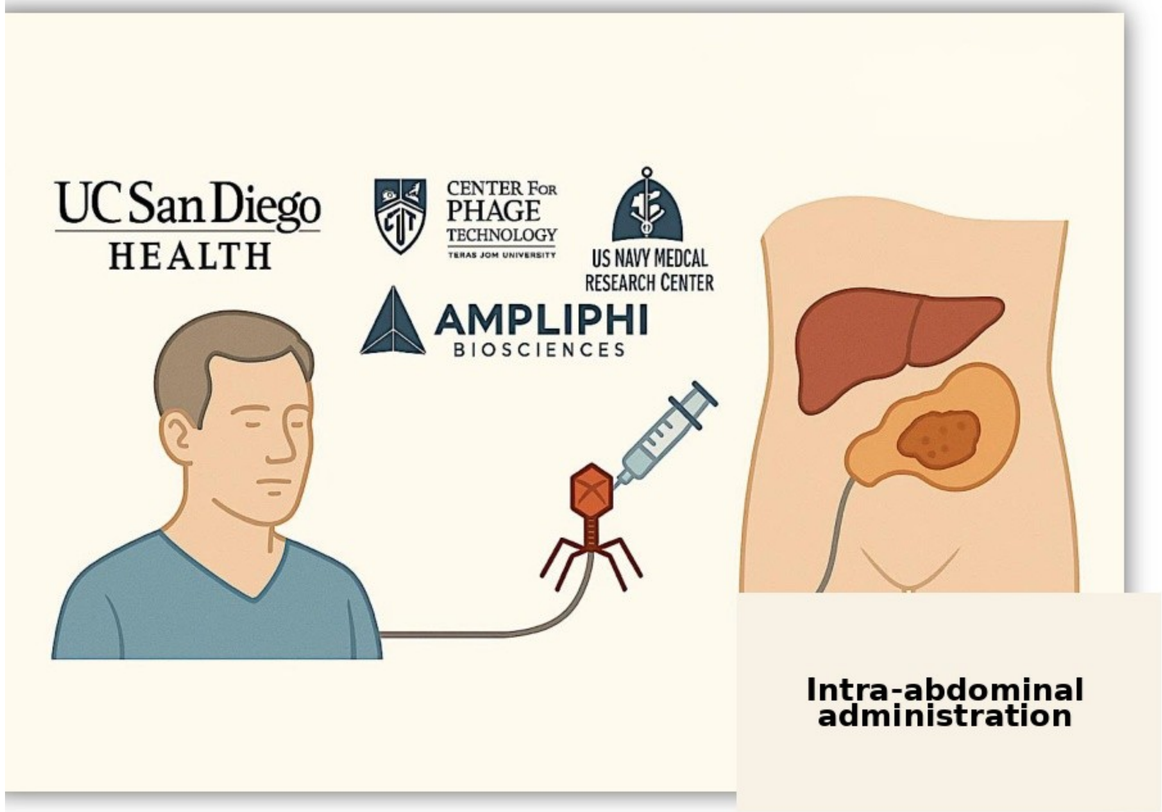


Figure 8. Schematic based on the clinical case

Source: Authors' own work

Innovative proposal

Several studies have observed that phage therapy can help regulate inflammatory processes, either through direct action of the pathogen or through other effects in which the organism is compromised. (Cui et al., 2024) (de Souza et al., 2023)

In this context, it is relevant to consider that eliminating microplastics from the organism could have a direct impact on the treatment of certain inflammatory or immune-mediated diseases. The presence of microplastics in the body has been associated with alterations in the microbiota, chronic inflammatory processes, metabolic disruptions, pulmonary diseases and even disorders affecting the nervous system. (Liu and Shimizu, 2025)

Based on the analysis of the immunological and microbiological effects of phage therapy, as well as the potential of chitosan as an adsorbent biopolymer, an interdisciplinary hypothesis is proposed: combining both strategies for advanced therapeutic purposes. Chitosan has shown a high capacity for microplastic adsorption and high biocompatibility with biological organisms. (Nicholas, 2024) (Prasetyo et al., 2025)

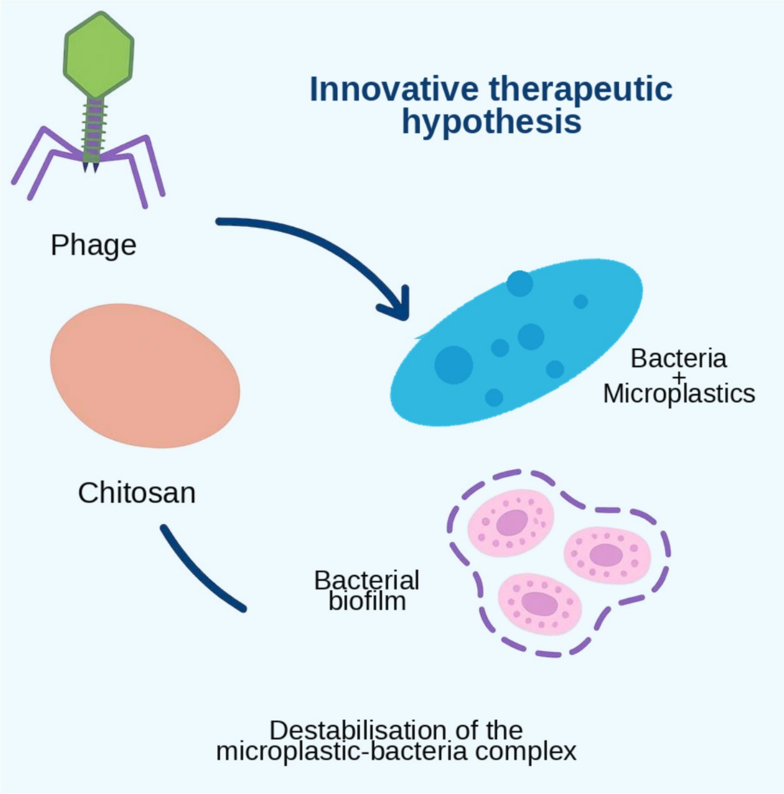


Figure 9. Schematic representation of the proposed therapeutic hypothesis for the elimination of microplastics using chitosan and phage therapy.

Source: Authors' own work

Conclusion

The potential of phages expands when they are combined with other therapies, such as antibiotics or lytic enzymes derived from the viruses themselves. This may enhance treatment efficacy and reduce the emergence of resistance.

However, despite scientific and clinical advances, the main obstacle to widespread implementation remains the lack of a specific legal framework that allows protocolised use. At present, application is mainly limited to compassionate or experimental treatments, restricting its reach within the healthcare system.

Thank you for your attention

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Coordinator: Javier Blanco Velazquez

This work was developed as a final programme project, integrating scientific knowledge with innovative personal contributions.

Bibliography

• microBIO. (2012, June 5). Bacteriófagos (virus que matan bacterias) en vez de antibióticos – microBIOblog. https://microbioblog.es/bacteriofagos-virus-que-matan-bacterias
• WHO. (2024, November 15). La crisis de resistencia a los antimicrobianos. https://news.un.org/es/story/2024/11/1534346
• Salud y Fármacos. (2022). Carga mundial de la resistencia bacteriana a los antimicrobianos en 2019: Un análisis sistemático. https://www.saludyfarmacos.org/lang/es/boletin-farmacos/boletines/may202205/24_ca
• d'Herelle, F. (with Internet Archive). (1926). The Bacteriophage: Its Role in Immunity. Williams & Wilkins. http://archive.org/details/bacteriophageits0000feli
• d'Herelle, F. (with MBLWHOI Library). (1922). The bacteriophage and its behavior. Baltimore, Md.: The Williams & Wilkins Company. http://archive.org/details/bacteriophageits00dher
• Bacteriófago: Características, partes y ciclo de vida. (2023, December 14). Tua Saúde. https://www.tuasaude.com/es/bacteriofago/
• Red de Recursos Educativos en Abierto. (n.d.). Retrieved May 10, 2025, from https://procomun.intef.es/gl/ode/view/es_2025032022_9143720
• da Rosa, J. E. C. (2015). Multirresistência bacteriana–uma "nova" terapêutica: Bacteriófagos.
• Melendez, M. (n.d.). Conoce la Biodiversidad y Propone como Preservarla. Retrieved May 19, 2025, from http://melendezcarmona.blogspot.com/2013/06/bloque-5-conoce-la-biodiversidad-y.html
• Vázquez, R., Díez-Martínez, R., Domingo-Calap, P., García, P., Gutiérrez, D., Muniesa, M., Ruiz-Ruigómez, M., Sanjuán, R., Tomás, M., Tormo-Mas, M. Á., and García, P. (2022). Essential Topics for the Regulatory Consideration of Phages as Clinically Valuable Therapeutic Agents: A Perspective from Spain. Microorganisms, 10(4), Article 4. https://doi.org/10.3390/microorganisms10040717

Bibliography

• Schooley, R. T., Biswas, B., Gill, J. J., Hernandez-Morales, A., Lancaster, J., Lessor, L., Barr, J. J., Reed, S. L., Rohwer, F., Benler, S., Segall, A. M., Taplitz, R., Smith, D. M., Kerr, K., Kumaraswamy, M., Nizet, V., Lin, L., McCauley, M. D., Strathdee, S. A., … Hamilton, T. (2017). Development and Use of Personalized Bacteriophage-Based Therapeutic Cocktails To Treat a Patient with a Disseminated Resistant Acinetobacter baumannii Infection. Antimicrobial Agents and Chemotherapy, 61(10), e00954-17. https://doi.org/10.1128/AAC.00954-17

• Lebeaux, D., Merabishvili, M., Caudron, E., Lannoy, D., Van Simaey, L., Duyvejonck, H., Guillemain, R., Thumerelle, C., Podglajen, I., Compain, F., Kassis, N., Mainardi, J.-L., Wittmann, J., Rohde, C., Pirnay, J.-P., Dufour, N., Vermeulen, S., Gansemans, Y., Van Nieuwerburgh, F., and Vaneechoutte, M. (2021). A Case of Phage Therapy against Pandrug-Resistant Achromobacter xylosoxidans in a 12-Year-Old Lung-Transplanted Cystic Fibrosis Patient. Viruses, 13(1), 60. https://doi.org/10.3390/v13010060

• Cui, L., Watanabe, S., Miyanaga, K., Kiga, K., Sasahara, T., Aiba, Y., Tan, X.-E., Veeranarayanan, S., Thitiananpakorn, K., Nguyen, H. M., and Wannigama, D. L. (2024). A Comprehensive Review on Phage Therapy and Phage-Based Drug Development. Antibiotics, 13(9), 870. https://doi.org/10.3390/antibiotics13090870

• de Souza, E. B., Pinto, A. R., and Fongaro, G. (2023). Bacteriophages as Potential Clinical Immune Modulators. Microorganisms, 11(9), 2222. https://doi.org/10.3390/microorganisms11092222

• Liu, D., and Shimizu, M. (2025). Ingesting chitosan can promote excretion of microplastics. Scientific Reports, 15, 14041. https://doi.org/10.1038/s41598-025-96393-w

• Nicholas, R. (n.d.). CHITOSAN-BASED HYDROGELS FOR EFFECTIVE MICROPLASTIC ADSORPTION IN WASTEWATER TREATMENT.

• Prasetyo, S., Santos, C. A., Sugih, A. K., and Kristianto, H. (2025). Utilization of chitosan as a natural coagulant for polyethylene microplastic removal. Sustainable Chemistry for the Environment, 9, 100225. https://doi.org/10.1016/j.scenv.2025.100225